PelagicLens: Ocean-aware fisheries signal reliability



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1. Comments

- I do not think the project should compete with another habitat map or another “where to fish” tool. The proposal should solve an interpretation problem: when the ocean changes the relationship between the true abundance of a pelagic stock and the signal observed by a fleet or a monitoring programme.

2. The problem we address

Abundance indices derived from commercial fisheries and scientific monitoring programmes, such as CPUE or survey-based indices, are a fundamental source of information on the temporal dynamics of fish stocks. Their interpretation as indices of relative abundance assumes, explicitly or implicitly, that the relationship between population state and the observation process remains sufficiently stable through time. In simplified form, an abundance index can be expressed as:

$I_{\square} = q_{\square} B_{\square} \exp(\varepsilon_{\square})$

where $I_{\square}$ is the observed index, $B_{\square}$ represents abundance or biomass, $q_{\square}$ is an effective observation coefficient, and $\varepsilon_{\square}$ represents observation error. Depending on the data source, $q_{\square}$ may integrate processes of availability, catchability, detectability, spatial coverage, and overlap between the population and the observation system.
For pelagic species, assuming that $q_{\square}$ remains constant can be particularly restrictive. Ocean variability can rapidly alter horizontal and vertical distribution, aggregation, the extent of occupied habitat, and proximity to the coast. These changes, in turn, alter the overlap of the population with scientific surveys or fishing fleets and, therefore, the fraction of the stock that is effectively observable or accessible at a given time. Consequently, $\Delta I_{\square}$ does not necessarily imply $\Delta B_{\square}$, because part of the variation in the index may originate from $\Delta q_{\square}$.

For example, spatial habitat compression may increase local density and encounter rates with the fleet, generating an increase in CPUE even when total biomass remains stable or decreases. Conversely, displacement of the stock outside the

usual survey domain could reduce the index without a proportional decline in population abundance.

The central problem is therefore one of identifiability between the population process and the observation process: what proportion of the observed variation in an index corresponds to changes in B□, and what proportion to changes in q□? Can oceanographic conditions and changes in the spatial distribution of pelagic fish explain and predict temporal variation in availability, detectability, or catchability, thereby identifying periods when abundance indices cease to be proportionally representative of stock status?
PelagicLens will address this problem by integrating Copernicus Marine oceanographic information, changes in fish distribution and spatial concentration, and the spatial footprint of the observation system to detect and anticipate conditions under which the relationship between the observed index and underlying abundance may become non-stationary.

3. Value proposition

PelagicLens transforms Copernicus Marine oceanographic information into intelligence for interpreting fisheries indicators under changing environmental conditions. The platform will integrate observed and forecast oceanographic conditions, spatiotemporal models of pelagic species distribution, and the spatial footprint of observation systems (scientific surveys, acoustics, fleets, or other sources) to estimate how environmental change modifies fish distribution, concentration, availability, and catchability. The objective is not only to predict where a species will occur, but to determine when a change in an index may be driven, wholly or partly, by changes in the observation process rather than by an equivalent change in stock abundance. Operationally, PelagicLens will convert the following chain:
OCEAN > FISH > OBSERVATION > SIGNAL > DECISION

- OCEAN represents environmental state and forecast derived from Copernicus Marine.
- FISH represents the spatial response of the population: displacement, habitat contraction or expansion, concentration, and redistribution.
- OBSERVATION represents the interaction between that distribution and the system attempting to observe or exploit it.
- SIGNAL evaluates how these processes may modify q□ and, therefore, the relationship between the observed index I□ and underlying abundance B□.
- DECISION translates this information into a diagnosis of index reliability and, when possible, an early warning based on Copernicus Marine forecasts.

4. Target users

- Stock assessment scientists who need to determine whether temporal variation in an index reflects changes in abundance or changes in availability, catchability, or detectability.
- Fisheries monitoring teams and scientific survey programmes, including acoustic surveys, that need to evaluate how fish redistribution may affect the representativeness of their observations.
- National fisheries agencies, research institutes, and regional fisheries management organisations that use abundance indices as inputs to stock assessment, scientific advice, and management decisions.
- Fisheries and organisations that depend on CPUE or other operational indicators, particularly where changes in aggregation, accessibility, or spatial distribution may alter the relationship between catch and abundance.
- At a later stage, consultancies, certification bodies, fishing companies, and organisations with their own monitoring systems that need to evaluate the reliability of their indicators.

5. Demonstration case and open data

PelagicLens will be developed using open and reproducible data, complemented where necessary with simulated data to demonstrate operational MVP functionalities.
The proposed initial case is European anchovy (Engraulis encrasicolus) in the Bay of Biscay / Iberian shelf. The main data sources will be:

- Copernicus Marine: historical and forecast physical and biogeochemical variables.
- ICES Acoustic Database / public surveys / DATRAS: structured observations, indices, spatial coverage, and sampling effort when available.
- OBIS: occurrence records for modelling potential distribution or environmental suitability.
- Simulated data: vessel trajectories, effort, or observation systems to test scenarios and demonstrate PelagicLens in near-real time.

6. Technical and scientific architecture of the MVP

6.1. OCEAN: Environmental state and forecast

Automated ingestion of historical and forecast physical and biogeochemical products from Copernicus Marine.
A reduced set of ecologically grounded variables will be prioritised:

- temperature and thermal anomalies; salinity; chlorophyll and/or primary production; mixed-layer depth; oxygen, where coverage is adequate; currents.

When they provide additional information, derived metrics will include:

- thermal gradients; frontal intensity; current speed; anomalies relative to a climatology; simple indicators of mesoscale activity.

Output
A spatiotemporal surface of ocean state, E(x,t), and, when forecasts are available, E(x,t+h).

Visual product
Map of current conditions + anomalies + temporal slider towards the forecast.

6.2. FISH: Spatial response of the population

PelagicLens will use a spatial or spatiotemporal model to relate biological observations to oceanographic conditions and reconstruct the dynamic distribution of the population. An interpretable statistical model will be prioritised and, as a contrast, a machine-learning algorithm may be incorporated to evaluate predictive robustness. The framework will not be restricted to a single type of Species Distribution Model. Depending on the available information, it may estimate presence, environmental suitability, relative density, or spatial biomass. The objective is not only to predict where the species may occur, but to characterise how the environment modifies its position, extent, concentration, aggregation, and spatial structure, and how those changes subsequently alter its availability to observation systems or fisheries.

D(x,t) = f{E(x,t), space, time}

where D(x,t) represents the spatial state of the resource and, depending on the available data, may correspond to probability of occurrence, relative occurrence intensity, density, or biomass. Candidate models may include GAM/GAMM, spatiotemporal models, delta/hurdle models, and machine-learning methods such as Random Forest or gradient boosting. Final model selection will depend on the observation type, validation capacity, and the need to produce reproducible and ecologically interpretable predictions (Mode 1-Habitat, Mode 2-Distribution, Mode 3-Density, Mode 4-Biomass).

Outputs
From D(x,t), PelagicLens will estimate a reduced set of ecological and spatial indicators:

- current and forecast spatial distribution;
- centre of gravity and direction of displacement;
- distance and speed of displacement;
- spatial extent and occupied area;
- habitat contraction or expansion;
- spatial concentration and aggregation;
- coastal–offshore displacement;

- changes in depth, where suitable data are available;
- spatial persistence and fragmentation;
- exposure to oceanographic anomalies;
- prediction uncertainty.

These indicators will provide the link between the ecological response of the population and the next PelagicLens layer.

Visual product

The application will display a dynamic map of population distribution, allowing comparison of the current state, a reference period, and, when possible, the Copernicus forecast.

The map will be accompanied by a compact indicator panel, for example:

Distribution shift: 63 km northward
Occupied area: −28%
Spatial concentration: +35%
Coastal displacement: −18 km
Habitat compression: HIGH
Forecast uncertainty: MODERATE

The user will also be able to switch between: Baseline → Current → Forecast.

6.3. OBSERVATION: Coverage and accessibility of the observation system

This layer will explicitly represent what fraction of the population distribution is potentially observable, detectable, or accessible to the system generating the index.

PelagicLens will integrate the estimated resource distribution with the geometry and intensity of the observation process, including, depending on data availability:

- scientific survey transects and stations;
- spatial coverage of acoustic surveys;
- fleet footprint;
- spatial distribution of effort;
- positions, speed, and trajectories of open or simulated vessels;
- sampling intensity and frequency.

We define O(x,t) as the spatial coverage or intensity of the observation system and D(x,t) as the estimated resource distribution. The central interaction is the overlap between both processes, D(x,t) ∩ O(x,t). An initial metric of effective coverage can be expressed as:

$$\Omega = \Sigma_x D(x,t)\, O(x,t) / \Sigma_x D(x,t)$$

where Ω represents the relative fraction of the resource distribution covered by or accessible to the observation system. The objective is not only to determine whether the fish and the survey/fleet coincide spatially, but to quantify how that coincidence changes through time and under different oceanographic conditions.

Priority outputs

- population–survey overlap;
- population–fleet overlap;
- relative proportion of the resource inside and outside the observed footprint;
- distance between the centre of resource distribution and the centre of the observation system;
- temporal changes in accessibility or coverage;
- spatial representativeness of sampling;
- areas with high probability of occurrence but low observation coverage;
- sensitivity of overlap to changes in distribution;
- uncertainty associated with predicted distribution and observation footprint.

This layer provides the link between the spatial ecology of the resource and the process that ultimately generates the fisheries index.

Visual product

The application will simultaneously display:

- the estimated resource distribution;
- observation transects, stations, or tracks;
- sampling or effort intensity;
- overlap zones and potentially unobserved areas.

The user will be able to switch between: FISH ONLY > OBSERVATION > OVERLAP and compare: Baseline → Current → Forecast.
The indicator panel could show, for example:
Survey overlap: 74%
Change from baseline: −18%
Unobserved high-density area: 22%
Distance fish–survey centroid: +46 km
Sampling representativeness: MODERATE
For a fleet:
Fleet overlap: +31%
Accessibility: HIGH
Habitat outside fishing footprint: 14%

6.4. SIGNAL: Ocean-informed interpretation of the index

Its function will be to integrate information generated by the OCEAN, FISH, and OBSERVATION layers to evaluate whether current environmental and spatial conditions may be modifying the relationship between the observed index and underlying abundance. PelagicLens will not assume that q□ is constant. Instead, it will evaluate whether changes in distribution, concentration, habitat compression, and overlap with the observation system are consistent with a deviation in q□ relative to reference conditions. Conceptually, q□ = f(E□,D□,O□). For the MVP, this relationship will be represented by a quantitative indicator of risk of change in

availability/catchability; when the data allow probabilistic calibration, it may be expressed as a probability. Subsequent scientific development will allow formal estimation of q□ and its uncertainty.

Priority outputs

The SIGNAL layer will synthesise the previous indicators into a reduced set of metrics:

- Habitat Compression: degree of contraction or expansion of the distribution relative to the baseline.
- Observation/Fleet Overlap: change in overlap between the population and the observation system.
- Availability Shift: estimated change in the fraction of the resource that is potentially observable or accessible.
- Catchability Shift Risk: risk that current conditions produce an important deviation in q□.
- Signal Reliability: degree of confidence with which the index can be interpreted as a proportional signal of abundance.
- Forecast Signal Risk: expected evolution of these metrics using Copernicus Marine forecasts.

Conceptually: E□ + D□ + O□ → P(|Δq□| > δ), where δ represents a change considered relevant relative to reference conditions.2

Interpretation

The main output will not simply indicate whether the index increases or decreases, but whether observation conditions remain comparable with the reference period. For example:

I□ ↑ + Habitat Compression ↑ + Fleet Overlap ↑

could indicate that part of the observed increase in CPUE is associated with greater concentration and accessibility of the resource.

Visual product

The interface will display a direct comparison between: RAW INDEX <-> OCEAN-AWARE INTERPRETATION, accompanied by a concise panel such as:
Observed index: +18%
Habitat compression: HIGH
Observation overlap: +26%
Catchability shift risk: HIGH
Signal reliability: LOW–MODERATE
Main driver: increased spatial concentration
And an automatic interpretation:
“The observed increase may be partially explained by greater concentration and accessibility of the stock. The index should therefore not be interpreted as a

proportional increase in abundance without accounting for changing observation conditions."

6.5. DECISION: Translating the diagnosis into decision support

The DECISION layer will transform SIGNAL results into information that can be directly interpreted by scientists and managers. Its objective is not to automate management decisions, but to indicate when an index may be affected by changes in availability, catchability, or observability and therefore requires more cautious interpretation.

SIGNAL → DIAGNOSIS → ALERT → INFORMED ACTION

PelagicLens will combine the magnitude of change in q□, index reliability, the main explanatory mechanisms, and associated uncertainty to generate a diagnosis.

Outputs

- Signal status: current condition of the indicator.
- Reliability level: reliability relative to the baseline.
- Dominant mechanism: main process associated with the observed change.
- Early-warning alert: advance warning when Copernicus forecasts indicate an upcoming change in observability or catchability.
- Decision context: guidance on how to interpret the index without replacing scientific judgement.

A simple classification for the MVP could be:

NORMAL

Observation conditions are similar to the reference period.

CAUTION

Moderate changes in distribution or overlap are detected and may affect index comparability.

HIGH OBSERVATION SHIFT

Environmental and spatial conditions suggest an important change in availability/catchability; the index should not be interpreted as proportional to abundance without accounting for this effect.

Visual product

The interface will display a final decision card:

Signal status: CAUTION

Reliability: Moderate

Catchability shift risk: High

Main driver: habitat compression + increased fleet overlap

Forecast: risk expected to increase over the next 72 h

accompanied by an automatic interpretation:

“Current ocean conditions are increasing the accessibility of the stock. Recent increases in the observed index may therefore overstate changes in underlying abundance.”
When a forecast is available:
“Observation conditions are expected to depart further from the historical baseline during the next 3 days.”

7. User experience: five screens

Each screen answers a specific question and naturally leads to the next.

Screen 1: OCEAN -What is happening in the environment?

Dynamic map using Copernicus Marine products and a temporal selector:

Baseline | Current | Forecast
It will show: current oceanographic conditions; anomalies relative to the reference period; temporal evolution; available forecast.
Main variables: temperature, chlorophyll/productivity, salinity, oxygen, currents, and relevant derived variables.
Visual product: animated map + time series + automatic identification of the main anomalies.

Screen 2: FISH -How is the population responding spatially?

It will show the estimated resource distribution under observed or forecast environmental conditions.
Main indicators: displacement of the distribution centre; occupied area; concentration/aggregation; compression or expansion; coastal–offshore or depth changes where relevant. Main comparison:
Baseline → Current → Forecast
Visual product: distribution map + displacement arrows + cards showing the main spatial changes.

Screen 3: OBSERVATION -What fraction of the population are we actually observing?

Overlay of: resource distribution; survey transects or stations; fleet footprint; effort; real or simulated telemetry. The user should immediately identify areas with:
high fish occurrence + high observation coverage
and areas with:
high fish occurrence + low observation coverage
Main indicators: survey/fleet overlap; relative coverage; population potentially outside the observed area; distance between the population and the observation system.
Visual product: overlap map and potentially under-observed areas.

Screen 4: SIGNAL -Did the stock change, or did our ability to observe it change?

This screen will integrate OCEAN + FISH + OBSERVATION.
It will show: Habitat Compression; Observation/Fleet Overlap; Availability Shift; Catchability Shift Risk; Signal Reliability; uncertainty.
Example:
Observed index: +18%
Habitat compression: HIGH
Fleet overlap: +27%
Catchability shift risk: HIGH

Interpretation
The observed increase could be partially associated with greater concentration and accessibility of the resource, and not necessarily with a proportional increase in biomass.

Screen 5: DECISION -How should the index be interpreted, and what may happen next?

The final screen will synthesise all information into an operational diagnosis.
It will show:
Signal status
NORMAL / CAUTION / HIGH OBSERVATION SHIFT
Reliability
HIGH / MODERATE / LOW
Main driver
Habitat compression + increased fleet overlap
Forecast
Observation conditions expected to deteriorate during the next 72 h.
Visual product: an executive decision card accompanied by a concise explanation of the main mechanisms generating the alert.

8. Provisional team roles

• Elmer - Scientific/Product Lead: scientific problem, pelagic ecology, ecological model, metric definition, validation, paper narrative, and scientific pitch.
• Tara - Ocean layer: selection/interpretation of Copernicus products, physical oceanography, and forecasts.
• Niklas - Data/Backend/Telemetry: Python, SQL, pipelines, APIs, mock telemetry, and ingestion architecture.
• Laia - Geospatial Analytics: grids, spatiotemporal joins, footprints, overlap, and geospatial visualisation.
• Abel - Full-stack/Product UI: dashboard, interaction, deployment, and user experience.